

SAMPLE DOCUMENT - FICTITIOUS DATA FOR DEMONSTRATION ONLY

Relief Load Summary - PSV-101

Process document RLS-101 Rev A. Relief study for separator V-101 (fictitious).

Tag No.	PSV-101
Protected equipment	V-101 HP Gas/Condensate Separator
P&ID	PID-100-002
Governing relief case	External fire
Fluid	Hydrocarbon vapour
Fluid state	Vapour
Required relieving capacity	18,500 kg/h
Molecular weight	19.6
Ratio of specific heats (Cp/Cv)	1.27
Compressibility factor Z	0.92
Operating temperature	85 °C
Relieving temperature	160 °C
Operating pressure	12 barg
Set pressure	15 barg
Allowable overpressure	21 % (fire case)
Superimposed back pressure	0.5 barg
Built-up back pressure	0.7 barg

Other cases checked: blocked outlet (8,200 kg/h) and gas blow-by (not credible, control valve sized below the separator design pressure).